Introduction to Computer Science

Mini project

Student workbook

makecode.microbit.org

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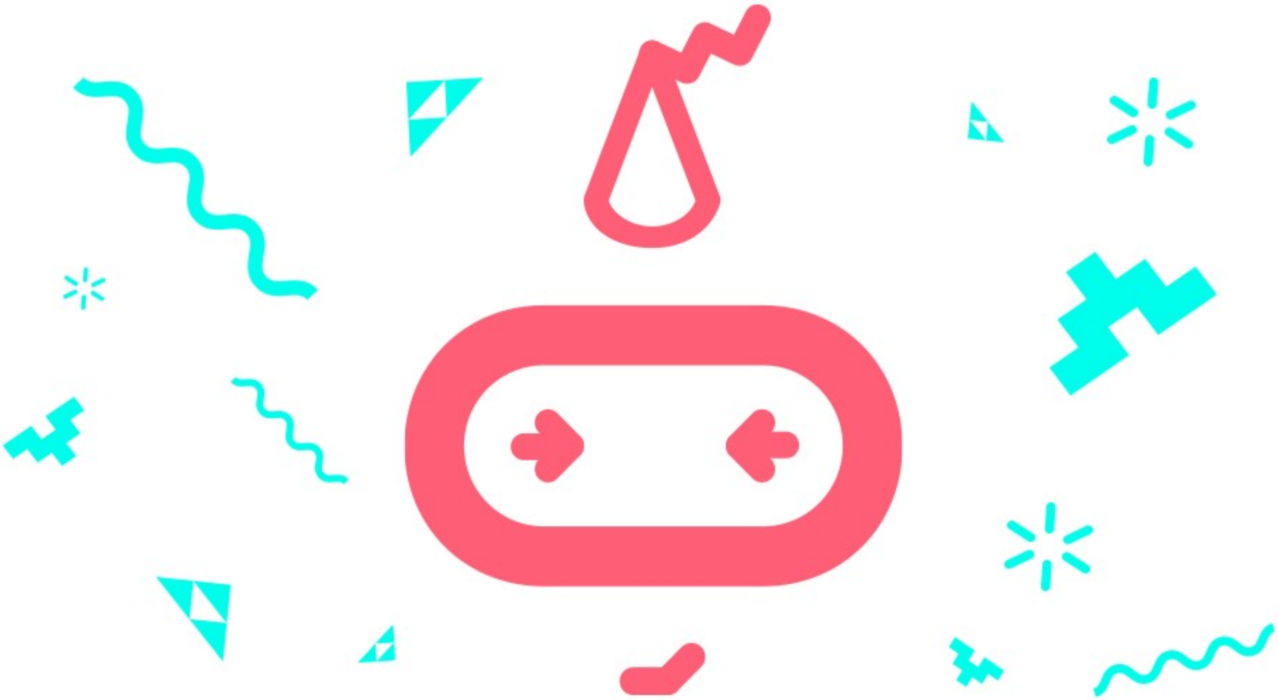


Overview

Unit summary

This this unit, you will be reviewing the concepts covered in the previous units and brainstorm some ideas for an independent "Mini project" that you will focus on in the next several classes. You will also be introduced to a framework for keeping yourself accountable to the work you are doing individually and in groups.

It is important to practice tracking the work you are doing on a short "Mini project" like this, so that when you move on to the final independent project, it will be easier for you to track your progress and work.

Tracking your work also reinforces the important idea that how you solve problems is at least as important to learning as whether you solved them at all (or even got the right answer). Programming is a process of patient problem solving, and finding ways to value, acknowledge, and reward the problem-solving process is an important part of assessment.

Learning goals

During this unit, you will:

- Code a unique, original program and design and build a physical maker component of the project that uses the micro:bit in some way.
- Demonstrate the use of one of the following concepts to illustrate what you know and show something new:
  - Input/processing/output
  - Variables
  - Simple Circuits
  - Iteration/loops
  - Conditional statements

Lesson A: Looking back so far

Overview: Mini project

You're at the midpoint of this course! It's a great time to reinforce what you've learned. In this unit, you'll start by reviewing the computer science concepts covered so far. Then, each of you will design and make a Mini project that serves a purpose by solving a problem or filling a need. It's an opportunity for each of you to do two things:

- Show what you know
- Demonstrate something new

Today's plan is to review what you've learned and brainstorm ideas for your Mini project. In the next lesson, we'll discuss the Mini project expectations and deliverables in more detail, and you'll start working on it.

Use this space to track your brainstorming ideas:

Lesson B: Coding and making a Mini project

Coding activity: Make and code a Mini project

The Mini project is an opportunity for you to design a project that serves a purpose by solving a problem or filling a need. It is also an opportunity to do two things:

- Show what you know
- Learn something new

Ideally, there should be a maker component to this project. This is a real-world component that works with the code on the micro:bit to do something unique.

You will be asked to use the design thinking approach to design, prototype, and beta test an original, independent project. You and your fellow students are allowed to work on the same idea, but you cannot turn in the same code as someone else. You can—and should—work collaboratively, solving the same kinds of problems together, but the projects should be unique and original.

Project expectations

Make sure your Mini project meets these specifications.

- Create an original project using the micro:bit.
- Incorporate a physical component to the project.
- Demonstrate the use of one of the following concepts:
  - Input/processing/output
  - Variables
  - Algorithms
  - Iteration/loops
  - Conditional statements
  - Making and design thinking

Project ideas

- New and improved fidget cube
- Moving monster
- Musical instrument
- Fishing game
- Air guitar Hhint: Use a ‘while’ loop to do tempo and pitch)
- Screensaver
- Screensaver that uses other inputs to draw
- Interactive book
- Binary clock or some other way to represent numbers visually

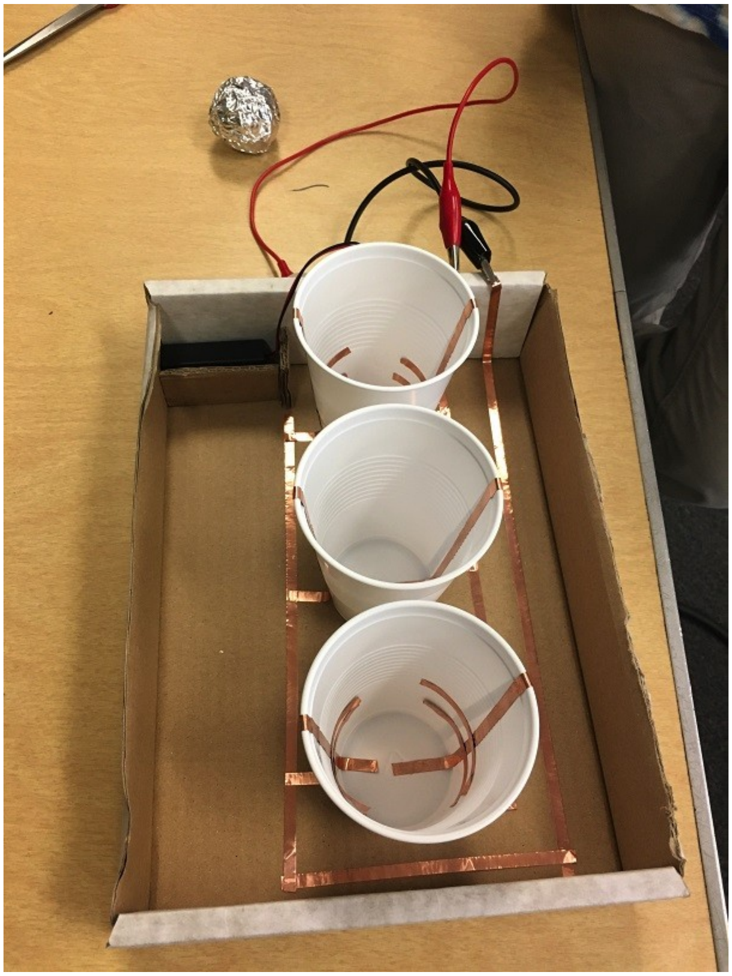
View projects at the following sites for inspiration:

- makecode.microbit.org/projects
- make.techwillsaveus.com/bbc-microbit
- microbit.org/ideas/
- twitter.com/MicroMonstersUK

Project examples

Toss the Ball

This is a skill game in which an aluminum foil ball is thrown into a plastic cup. Copper tape lining the sides and bottom of the cup completes the circuit when the ball touches it.

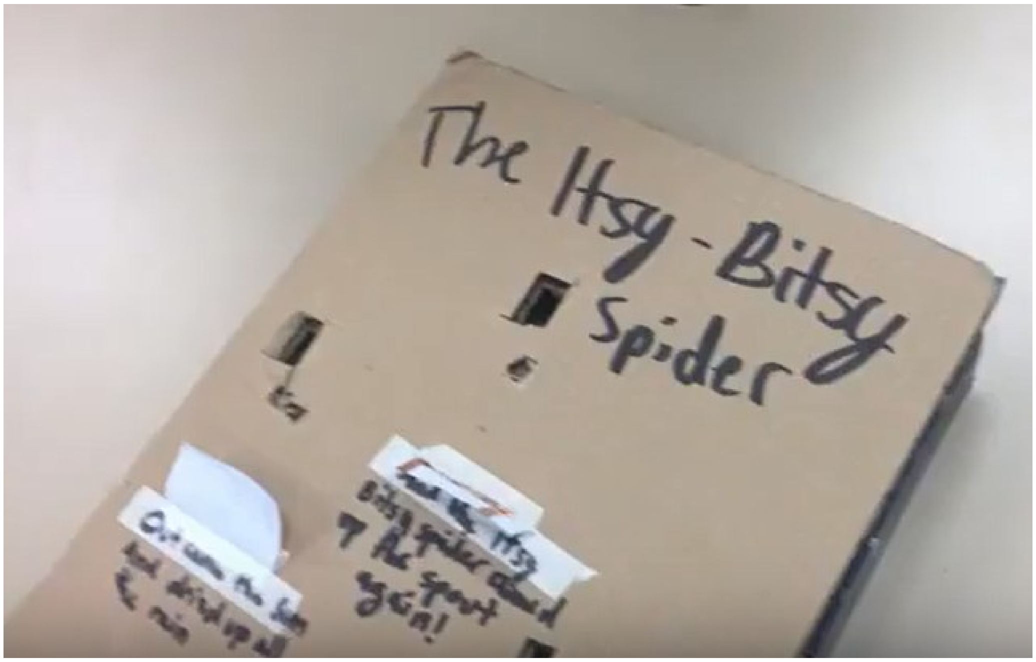


micro:bit bullseye

This is a skill game in which tennis balls are thrown underhand at one of the three rings, which are lined with aluminum foil so they complete a circuit underneath when the ball makes contact with the ring. See it in action at youtu.be/NZUpoSixf4E.

micro:bit storybook

This is a prototype of a storybook that could use the micro:bit to display animations for part of the story. Copper tape is used on the underside of the paper flaps to make contact between the GND pin and each of the other pins in sequence. See it in action at youtu.be/yg1NNLMqa9c (0:50)



Project scoring rubric

Assessment elements	1	2	3	4
Code: Show what you know	Code does not demonstrate previous concepts, is not efficient, variable names not clear.	Code only partially demonstrates previous concepts, and/or is not efficient, variable names not clear.	Code only partially demonstrates previous concepts, and/or is not efficient.	Code very effectively demonstrates the use of previous concept(s). Variable names are unique and clearly describe what information values the variables hold. Code is highly efficient.
Code: Show something new	Code does not demonstrate new concepts, is not efficient, variable names not clear.	Code only minimally demonstrates new concepts, and/or is not efficient, variable names not clear.	Code only minimally demonstrates new concepts, and/or is not efficient.	Code very effectively demonstrates the use of new concept(s). Variable names are unique and clearly describe what information values the variables hold. Code is highly efficient.
Maker component	No tangible component.	Tangible component does not add to the functionality of the program.	Tangible component is somewhat integrated with the micro:bit but is not essential.	Tangible component is tightly integrated with the micro:bit and each relies heavily on the other to make the project complete.

Work Logs

Expectations

A work log is a short, bullet point list of what you worked on and how long it took. You are encouraged to stick to the facts. It shouldn’t take more than thirty seconds or so to write up a work log. You should do one for every class.

Sample Work Log

April 11

- 20 min. Created code that reacts when pins P0 and P1 are pressed.
- 10 min. Talked with Mr. Kiang about how to attach wires so they won't fall off
- 20 min. Put target back together with pins
- 10 min. Helped Cody with attaching his scoreboard

Assessment elements	1	2	3	4
Work Logs	More than two late or missing work logs and/or not accurate nor sufficiently detailed.	Two late or missing work logs and/or work logs not accurate nor sufficiently detailed.	One late or missing work log and/or work logs not accurate nor sufficiently detailed.	All work logs submitted on time and complete.

The design thinking process:

1. Empathize by learning more about your target audience.
2. Define: understand and identify your audience's problems or needs.
3. Ideate: Brainstorm several possible creative solutions.
4. Prototype: Construct rough drafts or sketches of your ideas.
5. Test your prototype solutions and refine until you come up with the final version.

Use this space to start your design thinking process:

Lesson C: Mini project showcase

Reflection Diary

Expectations

At the end of the unit, you should compose a final reflection that summarizes the process of your learning over the course of the unit. You should go back through your Work Logs and reflect upon the following prompts:

- Talk about one challenge you faced in creating this project, either a challenge in coding or in making the artifact. How did you overcome this challenge?
- What did you demonstrate that you already knew?
- What was the new thing you learned in order to make this? How did you learn about it?
- Who in the class provided help to you along the way? How?
- Describe one specific thing you are proud of in this project.
- What would you do differently next time?
- If you had another week to work on this project, what might you add or improve?
- Publish your MakeCode project and include the URL.

Sample Reflection (excerpt)

I spent this week finishing up little details with my program, making it work better and more user friendly. The part that surprised me the most was the little things that kept popping into my head, little suggestions that could potentially be good to add, but might not be necessary or even useful. At the beginning of the assignment, I just added them as quickly as I thought of them, but as the project neared the midpoint and conclusion, I found myself considering if I actually needed them (as previous additions have since been quickly deleted).

Another thing that I find interesting about this is that it is a rather specialized project. Not many people would have a reason to use it except for me. However, this is supposed to be easily used by other people, so I have to take them into consideration as I design the project. I also realized that I had, at some point, broken part of my code without noticing it, so I now have to fix part of it. The reason that it is a problem is because I added a lot of code at once without testing it, which is unfortunate. Next time, I will add small amounts of code and test it first.

Diary entry scoring rubric

Assessment elements	1	2	3	4
Diary entry	Reflection piece lacks three of the required elements.	Reflection piece lacks two of the required elements.	Reflection piece lacks one of the required elements.	Reflection piece describes: ▪ Development Process ▪ Something new ▪ Something proud of ▪ Future mods